

assets. According to the NEO blockchain site:

Digital Assets + Digital Identity + Smart Contract = Smart Economy

Its consensus mechanism is based on Delegated Byzantine Fault Tolerance (dBFT), which supports a large volume of transactions per second. Randomly selected book-keeping nodes validate transactions on the network similar to Hyperledger Fabric and thus achieve consensus. It has evolved on the top of Ethereum's blockchain model, and its developers claim that it closes the loopholes that the Ethereum contracts had. NEO developers also claim to have built a security algorithm which is known to be quantum computing proof. Thus, present day quantum computing hacks cannot break NEO's algorithm.

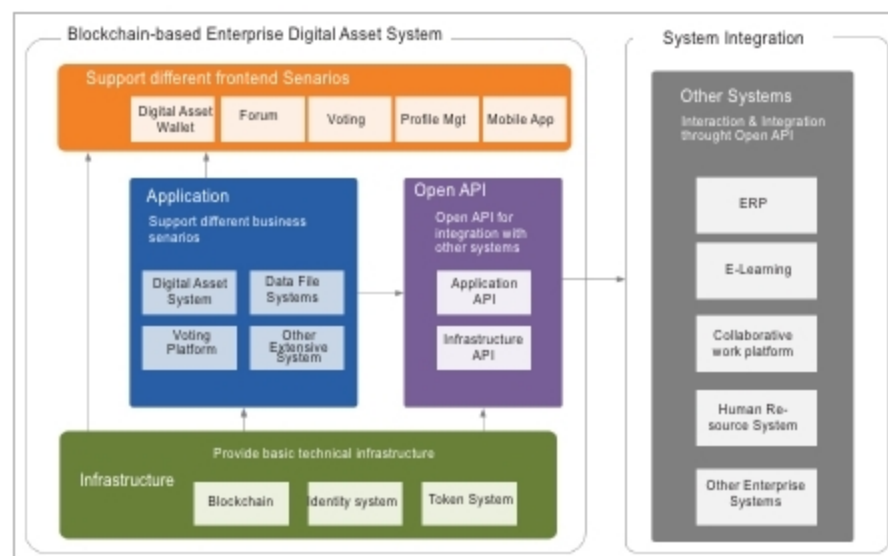


Figure 4: Neo blockchain ecosystem (Image source: <https://codeburst.io/all-about-neo-blockchain-ecosystem-you-ever-wanted-to-know-4f41fad09a78>)

NEO uses the NEOVM compiler to support several programming languages. It is crucial for network security and scalability that smart contract code gets isolated inside of the NEOVM. NEOVM is the lightweight implementation of Docker, which reduces the system resource requirements.

The following features differentiate NEO from other blockchain technologies:

1. NEP5 Communications Standard: This standard allows developers to build dApps using standardised workflows and templates.
2. X.509: NEO uses the X.509 digital identity standard, which is based on public key infrastructure. NEO also supports the Web of trust point-to-point certificate issuance mode. Identity verification is done with the use of facial features, fingerprints, voice, SMS, etc.
3. NeoX: The use of the NeoX protocol allows NEO tokens to communicate with tokens on other blockchains.
4. NeoFS: This is the InterPlanetary file system based decentralised file-sharing service.
5. NeoQ: This is the quantum-proof cryptographic mechanism that protects NEO from quantum

computing hacks.

NEO provides two types of digital assets:

1. Global assets
2. Contract assets

Smart contract development in the NEO blockchain:

Here is a sample of a smart contract developed using Java:

```
public class HelloWorld : SmartContract
{
    public static void Main()
    {
        Storage.Put(Storage.CurrentContext, "Hello",
"World");
    }
}
```

The storage class is a static class that manipulates the private contract storage. The *Storage.Put()* method allows you to store data in the private storage area in key-value format.

The Ethereum blockchain

The Ethereum blockchain structure is quite similar to the Bitcoin blockchain. Every node on the Ethereum network stores a copy of an entire transaction history. The recent state of each smart contract is stored on the nodes of the Ethereum blockchain, besides all other transactions.

There are two types of accounts on the Ethereum network:

1. External accounts
2. Contract accounts

Both accounts are considered as state objects and consist of the state of the Ethereum network. *Memory storage* and *balance* define the state of the contract account, while *account balance* defines the external accounts.

Ethereum uses a Proof-of-Work consensus mechanism, which is relatively old and used by the Ethereum and Bitcoin blockchains. It is quite slow and consumes much more power compared to the Proof-of-Stake consensus mechanism.

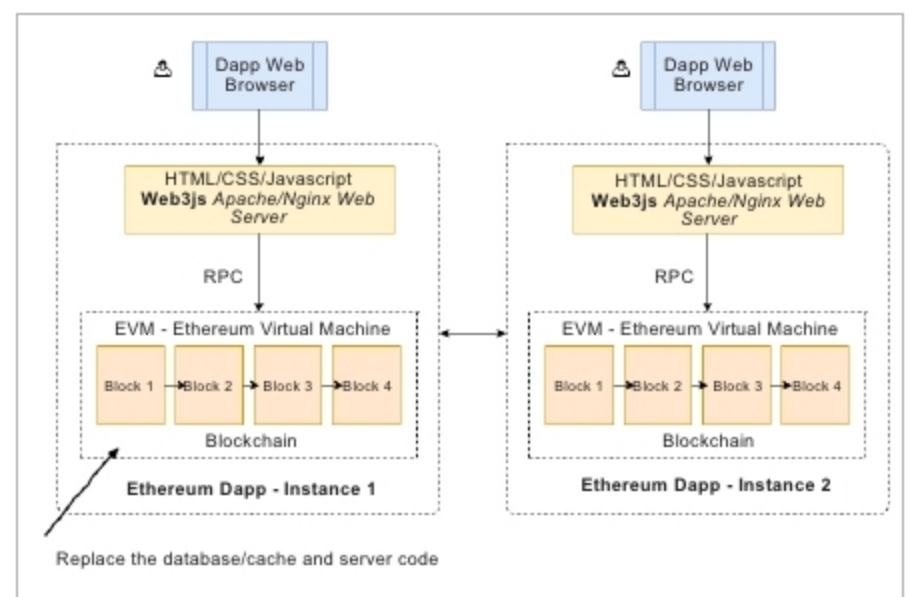


Figure 5: Ehtereum blockchain architecture (Image source: <https://www.zastrin.com/courses/ethereum-primer/lessons/1-5>)